Shared Mental Models in Generative AI-Based Multi-Agent Systems: A Simulation Study Grounded in Human Team Cognition Theory

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Abstract

GenAI-based multi-agent systems (MAS) enable coordinated problem-solving. However, the mere presence of multiple agents does not guarantee efficient and effective collaboration, especially when knowledge is distributed across agents. Drawing on shared mental model (SMM) theory, this study examines how an SMM-inspired structured memory mechanism influences the efficiency and effectiveness of GenAI-based MAS across different levels of access to prior discussion history. The analysis is based on 800 independent simulations of a hidden-profile personnel-selection task. A baseline without structured memory is compared with three structured memory configurations that provide agents with low, moderate, or high access to prior discussion history. The results show that the benefits of the structured memory mechanism depend on the context configuration. Under low access to prior discussion history, runs with the structured memory mechanism outperformed the baseline but required greater computational effort. This shows that computationally implementing mechanisms derived from human team cognition theory can provide a foundation for designing GenAI-based MAS.

Keywords: GenAI-based multi-agent systems, Shared mental models, Context management, Human team cognition theory, Multi-agent collaboration

GenAI-basierte Multi-Agenten-Systeme (MAS) ermöglichen das koordinierte Lösen von Problemen. Das bloße Vorhandensein mehrerer Agenten garantiert jedoch keine effiziente und effektive Kollaboration, insbesondere wenn Wissen über mehrere Agenten verteilt ist. Aufbauend auf der Theorie des Shared Mental Model (SMM) untersucht diese Studie, wie ein SMM-inspirierter strukturierter Gedächtnismechanismus die Effizienz und Effektivität GenAI-basierter MAS bei unterschiedlichem Zugang zum bisherigen Diskussionsverlauf beeinflusst. Die Analyse basiert auf 800 simulierten Personalauswahldiskussionen. Eine Baseline ohne strukturiertes Gedächtnis wird mit drei Konfigurationen mit strukturiertem Gedächtnis verglichen, die den Agenten einen geringen, moderaten oder hohen Zugang zum bisherigen Diskussionsverlauf gewähren. Die Ergebnisse zeigen, dass die Vorteile des strukturierten Gedächtnismechanismus vom jeweiligen Kontext abhängen. In Kombination mit einem geringen Zugang zum Diskussionsverlauf übertraf er die Baseline, war jedoch mit einem höheren Rechenaufwand verbunden. Das zeigt, dass die technische Umsetzung von Mechanismen aus der Theorie menschlicher Teamkognition eine Grundlage für die Gestaltung GenAI-basierter MAS schaffen kann.

Schlüsselwörter: GenAI-basierte Multi-Agenten-Systeme, Shared Mental Models, Kontextmanagement, Theorie menschlicher Teamkognition, Multi-Agenten-Kollaboration

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List of Abbreviations

ADK	Agent Development Kit
AI	Artificial Intelligence
API	Application Programming Interface
CI	Confidence Interval
GenAI	Generative Artificial Intelligence
HC3	Heteroskedasticity-Consistent Type 3
HR	Human Resources
I-P-O	Input-Process-Outcome
LLM	Large Language Model
MAS	Multi-Agent System
OLS	Ordinary Least Squares
SD	Standard Deviation
SE	Standard Error
SMM	Shared Mental Model

List of Symbols

$x_{1:n}$	Input context as a sequence of tokens
x_i	Individual token in the input context
$y_{1:m}$	Generated response as a sequence of tokens
y_j	Individual generated token
v	Possible next token
$\mathcal{V}$	Vocabulary of the language model
$z_{\theta,v}$	Logit assigned to token v by the model
θ	Model parameters
p_θ	Model-implied conditional probability distribution parameterised by θ
$\beta_j^{(m)}$	Regression coefficient j in model m
ε_i	Error term in a regression model
α	Significance level
p	p-value
R^2	Coefficient of determination
M	Mean
f^2	Effect size
$1-\beta$	Statistical power

1 Introduction

Imagine a scenario in which a company conducts an assessment where several applicants are evaluated across different exercises and interviews. Before a hiring decision can be made, the evaluators need to review, compare, and integrate their observations against the requirements of the role. Such decisions require decision makers to pool different pieces of information before arriving at a final choice. This scenario can be understood as complex knowledge work. Knowledge work is a form of labour in which the majority of value creation stems from the generation, application, and sharing of knowledge (Schultze, 2000). It can be considered complex when it involves non-routine and non-rule-based tasks rather than solely routine and repetitive tasks (Autor et al., 2003). Recent advances in generative artificial intelligence (GenAI) make it plausible that complex knowledge work can be supported, partially automated or fully performed by computational systems (Dell'Acqua et al., 2026). For text-based knowledge work, large language models (LLMs) are particularly relevant because they can generate and interpret natural language (Feuerriegel et al., 2024). However, complex forms of knowledge work involve several interdependent subtasks rather than a single homogeneous activity. While these could be addressed by a single LLM, they can also be distributed across multiple specialised LLMs that each focus on one subtask.

This shift leads to the concept of multi-agent systems (MAS), in which several autonomous agents dynamically coordinate to address complex, context-rich tasks within a shared environment (Allmendinger et al., 2026; Russell & Norvig, 2021; M. J. Wooldridge, 2009). Such systems consist of GenAI-based agents that use an LLM as their core reasoning and communication component. Each agent is assigned a specific role, goal, and information access (X. Li et al., 2024). MAS are particularly relevant for complex knowledge work, which often requires the integration of diverse cognitive skills and is therefore performed collaboratively (Pettersen, 2019). In such collective endeavours, agents interact and iteratively build on one another's contributions. In doing so, they jointly advance a complex task in small steps rather than aggregating independent outputs. Through such interdependent collaboration, multiple agents can accomplish tasks that exceed the capabilities of a single agent (Fetzer et al., 2026). However, if agents operate on inconsistent assumptions or incomplete task representations, the advantages of specialisation and collaboration are reduced (Tambe, 1997). Further, contemporary MAS rapidly converge through "affirmation-first" dynamics, bypassing critical elaboration (Knop et al., 2026). Thus, the mere presence of multiple agents does not guarantee effective and efficient collaboration.

This is consistent with research on human teams, which emphasises that collective performance depends not only on individual member resources, but also on team processes through which members coordinate their actions and combine their resources (Marks et al., 2001). In particular, the concept of the shared mental model (SMM) has been widely used to explain how shared understandings of the task, the team, and the interaction process enable efficient and effective collaboration and thereby improve team performance (Cannon-Bowers et al., 1993; DeChurch & Mesmer-Magnus, 2010; Mathieu et al., 2000).

Applying the concept of SMMs to artificial agents and human-agent teams is not entirely new (Scheutz et al., 2017; Yen et al., 2006). However, this line of work has not yet been transferred to contemporary GenAI-based MAS. Recent research has mainly examined the architectural and procedural organisation of agents, including specialised roles and subtasks, inter-agent communication structures, and the orchestration of task workflows (X. Li et al., 2024; Tran et al., 2025). While these studies show that MAS can support collaborative problem solving, current systems still struggle to integrate distributed information into correct group decisions (Y. Li et al., 2025). Thus, prior research points to an important gap: although SMMs are well established as an explanation for effective and efficient collaboration in human teams, it remains unclear whether an SMM-inspired structured memory mechanism can likewise improve the efficiency and effectiveness of GenAI-based MAS. Because GenAI-based agents generate their responses based on the information available in their input context (Brown et al., 2020), the effects of such a structured memory mechanism may also depend on how much prior discussion history agents can access. Accordingly, this study poses the following research question:

How does an SMM-inspired structured memory mechanism affect the efficiency and effectiveness of GenAI-based MAS across different levels of access to prior discussion history?

The research question is examined through a personnel-selection task adapted from Schulz-Hardt et al. (2006) in which three GenAI-based agents must jointly identify the most suitable candidate for a long-distance pilot position. Using Google's Agent Development Kit (ADK) framework, this study simulates 800 group discussions. A baseline without structured memory is compared with three structured memory configurations that provide agents with low, moderate, or high access to prior discussion history. By analysing communication quality, cooperation quality, efficiency, and effectiveness, this work examines whether computationally implementing mechanisms derived from human team cognition theory can provide a foundation for designing GenAI-based MAS.

2 Theoretical Background

2.1 GenAI-Based Multi-Agent Systems

Research on artificial intelligence (AI) has long been concerned with the design of systems that can perceive their environment, process information, and act towards goals. In this sense, AI systems are often conceptualised as agents (Russell & Norvig, 2021). Earlier AI agents were designed for well-defined, rule-based tasks and specific environments, such as chess-playing systems (Campbell et al., 2002). However, these systems were limited in their transferability. The emergence of GenAI, and particularly LLMs, has expanded the traditional understanding of AI agents. LLMs are large-scale, typically transformer-based neural networks with billions of parameters that are pre-trained on large amounts of text to interpret input and generate natural language output (Feuerriegel et al., 2024). Unlike earlier, more task-specific AI systems, they can perform a broad range of tasks by conditioning their outputs on prompts. A prompt is a textual input constructed to guide an LLM towards a desired response or action (X. Liu et al., 2026). This ability is known as in-context learning, where task behaviour is shaped by information made available at inference time rather than by changing the model parameters θ (Brown et al., 2020). Building on these capabilities, an LLM can serve as the core reasoning and communication component of an agentic system. Such a system is referred to as a GenAI-based agent and extends the LLM with additional components such as role descriptions, task goals, memory, tools, planning routines, and interaction rules (Wang et al., 2024). At each inference step, the information supplied by these components forms the agent's context. This context can be represented as a sequence of tokens:

$$x_{1:n} = (x_1, \dots, x_n)$$

Tokens are small text units, such as complete words, parts of words, punctuation marks, or numbers. For every possible next token v in the model's vocabulary $\mathcal{V}$, the LLM calculates a score, called a logit, $z_{\theta,v}(x_{1:n})$. These scores are converted into probabilities through the softmax function:

$$p_\theta(x_{n+1} = v \mid x_{1:n}) = \frac{\exp\left(z_{\theta,v}(x_{1:n})\right)}{\sum_{u \in \mathcal{V}} \exp\left(z_{\theta,u}(x_{1:n})\right)}$$

Put simply, the model assigns a probability to every possible next token, conditional on all information currently contained in the context. A response is not generated at once. Instead, the model produces one token, adds it to the existing text, and then predicts the following token.

Depending on the decoding procedure, the next token is either selected as the most probable token or sampled from the probability distribution. In either case, a previously generated token becomes part of the context for the following generation steps and therefore influences the remainder of the response. The probability of a complete response $y_{1:m}$ can therefore be represented as:

$$p_\theta(y_{1:m} \mid x_{1:n}) = \prod_{j=1}^{m} p_\theta\left(y_j \mid x_{1:n}, y_{1:j-1}\right)$$

Importantly, context does not modify the model parameters θ. Instead, it changes the textual input tokens $x_{1:n}$ on which the model bases its next-token predictions. Through the transformer architecture, these input tokens change the logits calculated for possible next tokens, and consequently change the generated response (Brown et al., 2020; Vaswani et al., 2017). Because different context configurations can therefore produce different responses from the same model, context management is a central design issue in GenAI-based agents. Context management can be understood as the process through which an agentic system controls what information is available to the model in subsequent calls (Mei et al., 2025). However, the amount of context that can be provided to the model is constrained by its context window, which limits the number of tokens that can be processed during a single inference step (Wang et al., 2024).

One mechanism of context management is memory. It enables an agent to preserve information across model calls and selectively reintroduce it into later contexts when needed (Zhang et al., 2025). In GenAI-based agents, memory is realised through external systems, like natural-language memory records, human-readable Markdown files, relational databases, or vector stores (Nguyen et al., 2026; Zhang et al., 2025). Memory mechanisms involve several related operations. Information must first be selected and written into memory. It must then be updated, combined, or removed. When the agent faces a new decision, relevant memory content must be retrieved and incorporated into the current context to influence the response generation (Park et al., 2023; Wang et al., 2024).

Memory becomes especially relevant when several GenAI-based agents interact within a shared environment. In such settings, each agent operates with its own role, goals, context, and memory, while also receiving information from other agents through interaction and coordinating its activities with those of other agents (Allmendinger et al., 2026; Wang et al., 2024). The resulting system is a MAS. It consists of multiple agents, where a single agent is a computer system characterised by two key properties. First, it can execute autonomous actions to achieve specified goals. Second, it is capable of interacting with other agents in ways that enable cooperation, coordination, and negotiation to solve problems that are difficult for a single agent to address alone (Russell & Norvig, 2021; M. J. Wooldridge, 2009). In GenAI-based MAS, interaction is often organised through natural-language communication among agents. Prior work on MAS shows that interaction among GenAI-based agents can improve reasoning and factuality (Du et al., 2024). However, these benefits are not automatic. MAS do not consistently outperform single-agent prompting approaches and depend on how the interaction is designed (Smit et al., 2024). Agents must exchange useful information and use it to reassess weak or incorrect answers rather than merely generate multiple independent responses (Wu et al., 2025). Thus, GenAI-based MAS face a context management problem in which agents must maintain a sufficiently aligned understanding of the evolving task, available information, and each other's contributions. This challenge is well established in research on human teams.

2.2 Shared Mental Models as Input to Team Processes

Research on human teams has proposed several theoretical models of team functioning (Guzzo & Dickson, 1996; Ilgen et al., 2005). Although such models differ in their specific details, they generally follow the input-process-outcome (I-P-O) framework (McGrath, 1964). Inputs are conditions that exist before a performance episode, including characteristics of individual members, the team, and the organisational environment. During a performance episode, in which team members engage in task-related work, team processes describe how these inputs are transformed into outputs. Outcomes comprise the results and by-products of team activity (Mathieu et al., 2000). Later extensions emphasise that teams engage in successive performance episodes, where feedback and information generated in one episode shape the inputs, team processes and outcomes of subsequent episodes (Ilgen et al., 2005; Mathieu et al., 2008). There are three broad types of outcomes: team performance, team longevity, and team members' affective reactions (Hackman, 1990). This research focuses on team performance because team longevity and team members' affective reactions are less applicable to short-lived MAS.

Following the I-P-O framework, explaining differences in team performance requires identifying the input factors that shape team processes during a performance episode. One central input factor is the SMM. A mental model is a simplified and incomplete cognitive representation that individuals use to interpret a situation, anticipate developments, and guide their actions (Rouse & Morris, 1986). In teams, SMMs refer to sufficiently compatible knowledge structures that enable members to form aligned expectations about the task, the team, and each member's contribution without requiring identical knowledge. This shared understanding allows team members to coordinate their actions, anticipate one another's information needs, and adapt their behaviour to changing task demands (Cannon-Bowers et al., 1993). For example, members of a firefighting team do not need to explicitly communicate every action during an emergency. Because they share knowledge of roles, procedures, risks, and the evolving situation, they can anticipate what others are likely to do and coordinate their own actions accordingly. Depending on the content of knowledge shared among team members, the literature distinguishes two types of SMMs: task mental models, which concern shared knowledge about the task, and team mental models, which concern shared knowledge about the team and its interaction processes (Mathieu et al., 2000).

Within the I-P-O framework, SMMs are expected to influence team performance by shaping team processes such as communication, cooperation, and coordination (Klimoski & Mohammed, 1994; Mathieu et al., 2000). Coordination refers to the alignment of the sequence and timing of interdependent actions so that individual contributions fit together in pursuit of a shared goal (Marks et al., 2001). Although some coordination may occur in the simulated MAS examined in this study, it is not analysed as a separate team process dimension because key coordination features are fixed by design.

Figure 1 illustrates how the functional role of SMMs can be transferred to GenAI-based MAS within the I-P-O framework.

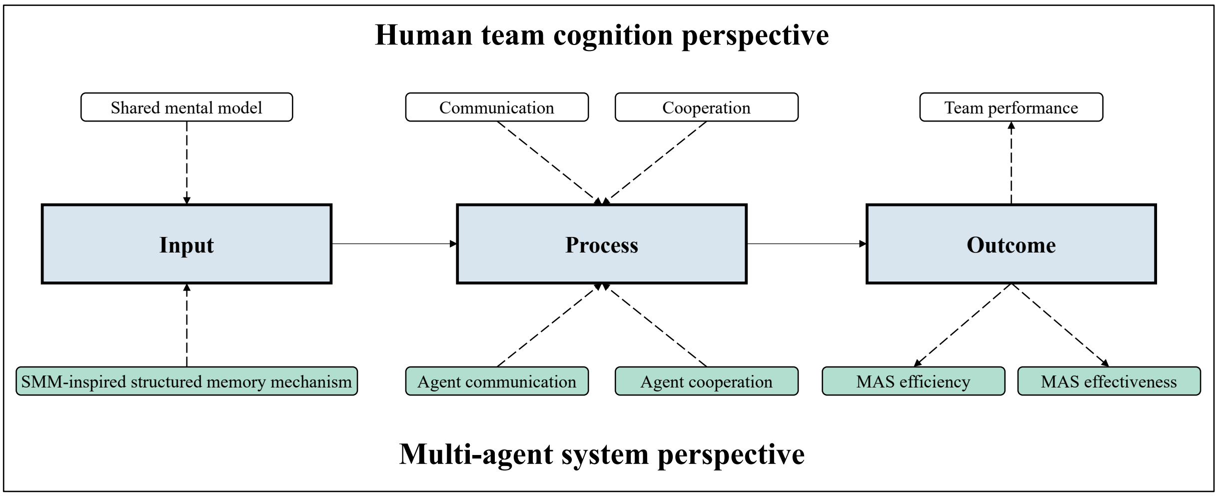

Figure 1: Mapping the human team cognition perspective onto GenAI-based MAS
Source: Own research (based on Mathieu et al. (2008), p. 413)

An SMM-inspired structured memory mechanism serves as a computational analogue of the functions associated with task and team mental models by preserving, updating, and repeatedly providing a structured representation of the team's evolving knowledge state. In this study, the structured memory records candidate-specific strengths and concerns, the discussion coverage, including relevant facts not yet discussed, the distribution of information across agents, agents' current positions, and unresolved issues affecting the group decision. Before an agent generates a subsequent contribution, the structured memory content is updated and reintroduced into its input context. It thereby emphasises the exchange, discussion, and integration of decision-relevant information. The resulting team performance can then be decomposed into efficiency and effectiveness (Tangen, 2005).

3 Hypotheses Development

In GenAI-based MAS, information is often distributed asymmetrically across specialised agents through different roles, instructions, and especially information access. Consequently, no single agent necessarily holds a complete representation of the task (X. Li et al., 2024). Effective MAS performance therefore depends on two requirements: whether initially distributed information is communicated and taken up by other agents and whether agents actively engage with and integrate one another's contributions into the collective assessment.

Communication quality concerns the first of these requirements. In distributed-information settings, communication allows initially dispersed knowledge to become available to the group, yet groups often focus more strongly on shared than on uniquely held information (Marlow et al., 2018; Mesmer-Magnus & DeChurch, 2009; Stasser & Titus, 1985). The structured memory mechanism should reduce this problem through two related mechanisms. First, it highlights what has already been discussed and which information gaps remain, thereby increasing the likelihood that agents disclose relevant private information. Second, once information has been disclosed, the structured memory persistently records it and makes it available in subsequent contexts, increasing the likelihood that it is taken up by another agent. This expectation is consistent with evidence that making the distribution and content of information visible can support the retention and use of unique information once it has been mentioned (Stasser et al., 2000) and that shared external representations can make relevant categories, relations, and information gaps salient to group members (Suthers, 2001).

Cooperation quality concerns whether agents actively engage with and integrate the information and assessments contributed by others rather than producing separate individual judgements (Van Knippenberg et al., 2004). In GenAI-based MAS, such cooperation becomes observable through information elaboration: the discussion, comparison, refinement, and integration of information into an emerging collective assessment (Van Knippenberg et al., 2004). A structured memory mechanism should support this process by providing a persistent representation of the evolving group knowledge state. Consistent with SMM theory, compatible task and team representations help team members recognise which information is relevant for the collective decision and how their own contributions relate to the group's current understanding (Cannon-Bowers et al., 1993). Similarly, shared representations that emphasise exchange, discussion, and integration of decision-relevant information have been shown to increase information elaboration (Van Ginkel & Van Knippenberg, 2008). The structured memory mechanism is therefore expected to improve communication quality by supporting both the disclosure and subsequent cross-agent uptake of initially private information, and cooperation quality by facilitating the integration of others' contributions into a collective assessment.

Hypothesis 1a: *GenAI-based MAS employing an SMM-inspired structured memory mechanism exhibit higher communication quality than systems without such a mechanism.*

Hypothesis 1b: *GenAI-based MAS employing an SMM-inspired structured memory mechanism exhibit higher cooperation quality than systems without such a mechanism.*

However, the effects of the structured memory mechanism should depend on the information available to the agents at inference time, which this study refers to as context configuration. It is operationalised as a four-level categorical variable comprising the baseline and three structured memory configurations. Each of the three configurations includes the structured memory mechanism but differs in the amount of prior discussion history available to the agents. These amounts correspond to low, moderate, and high levels of what this study terms discussion history transparency, occasionally shortened to transparency.

Discussion history transparency is expected to involve a trade-off between providing sufficient context to interpret earlier contributions and limiting the amount of information agents must process. Under low transparency, agents can access structured memory entries but receive little information about the reasoning from which they emerged. This may make previously disclosed facts and contributions less likely to be recognised, taken up, and integrated into subsequent assessments. Moderate transparency should alleviate this problem by supplementing structured memory with a reviewable discussion record. Access to this context supports grounding by enabling agents to establish sufficient mutual understanding of earlier contributions before extending the discussion (Clark & Brennan, 1991). Related research on long-term dialogue systems also indicates that access to relevant earlier exchanges improves the use of prior interaction compared with more limited-context approaches (Xu et al., 2022).

However, high transparency may expose agents to more discussion history than they can effectively process. Although additional information can initially support decision-making, excessive information volume can impede its effective processing (Mullins & Sabherwal, 2022). LLMs are particularly susceptible to distraction from irrelevant context, reduced use of information embedded in long inputs, and performance degradation as context length increases (Du et al., 2025; N. F. Liu et al., 2024; Shi et al., 2023). As a result, decision-relevant facts and contributions may become less salient and less likely to be taken up, compared, and integrated. Moderate transparency is therefore expected to provide the most favourable balance and produce higher communication and cooperation quality than both low and high transparency.

Hypothesis 2a: *Among the structured memory configurations, moderate discussion history transparency results in higher communication quality than low discussion history transparency.*

Hypothesis 2b: *Among the structured memory configurations, moderate discussion history transparency results in higher cooperation quality than low discussion history transparency.*

Hypothesis 3a: *Among the structured memory configurations, moderate discussion history transparency results in higher communication quality than high discussion history transparency.*

Hypothesis 3b: *Among the structured memory configurations, moderate discussion history transparency results in higher cooperation quality than high discussion history transparency.*

Within the I-P-O framework, team processes link inputs to team performance, which can be decomposed into efficiency and effectiveness (Tangen, 2005). Efficiency refers to the extent to which the MAS reaches a collective decision with minimal computational resources. Effectiveness refers to the degree to which the MAS achieves the correct collective task outcome.

Communication quality is expected to improve efficiency, reflected in lower token usage. Although every additional message creates computational cost, communication frequency alone does not determine efficiency. Rather, efficiency depends on whether decision-relevant information is disclosed and taken up by other agents. Such uptake allows agents to build on the existing knowledge state instead of repeatedly searching for and disclosing the same information. The resulting reduction in redundant processing may outweigh the computational cost of the interaction itself. This reasoning is consistent with evidence that task-relevant knowledge exchange is more strongly associated with team performance than communication frequency (Marlow et al., 2018).

Cooperation quality should further improve efficiency by ensuring that agents explicitly relate their own assessments to those contributed by others rather than producing isolated judgements (Dennis, 1996). By comparing, reconciling, and building on prior assessments, agents can resolve disagreements more directly and coordinate their decision making. This should facilitate consensus and enable agents to converge with fewer unnecessary messages and interaction rounds (Gu et al., 2012). Communication and cooperation quality are therefore expected to increase efficiency by reducing the computational effort required to reach a collective decision.

Hypothesis 4a: *In GenAI-based MAS, communication quality positively affects efficiency.*

Hypothesis 4b: *In GenAI-based MAS, cooperation quality positively affects efficiency.*

In addition, communication quality is expected to improve effectiveness as well. Because task-relevant information is distributed across agents, communication quality should increase the likelihood that uniquely held evidence is disclosed and taken up by the group. This is particularly important in distributed-information tasks, in which groups are more likely to reach the correct decision when relevant unique information is pooled during discussion (Lu et al., 2012). Additionally, information sharing is positively related to team performance, including decision effectiveness (Mesmer-Magnus & DeChurch, 2009).

However, information availability alone is not sufficient. Evidence from group support systems shows that although greater information exchange provides the informational basis for a correct decision, groups may still fail to identify the optimal alternative (Dennis, 1996). Cooperation quality should therefore improve effectiveness by enabling agents to compare, interpret, and integrate the available information into a coherent assessment. Consistent with this reasoning, shared task representations have been shown to improve decision quality through information elaboration (Van Ginkel & Van Knippenberg, 2008). Communication quality is thus expected to improve effectiveness by making distributed information available, whereas cooperation quality should improve effectiveness by ensuring that this information is jointly evaluated and used to select the correct alternative.

Hypothesis 5a: *In GenAI-based MAS, communication quality positively affects effectiveness.*

Hypothesis 5b: *In GenAI-based MAS, cooperation quality positively affects effectiveness.*

Figure 2 summarises the constructs examined in this study and their hypothesised relationships.

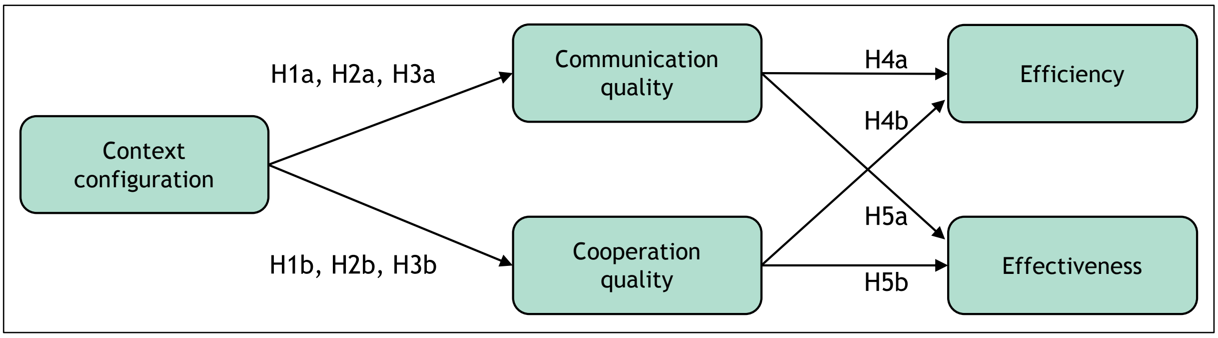

Figure 2: Conceptual research model and hypothesised relationships
Source: Own research

4 Methodology

All simulation runs used a self-hosted gpt-oss-120b open-weight model with temperature set to 0. Further details about the system configuration are reported in Appendix B.

The simulation task is a personnel-selection task adapted from Schulz-Hardt et al. (2006), which is based on the hidden-profile paradigm (Stasser & Titus, 1985). In the hidden profile paradigm, some decision-relevant information is shared by all group members, whereas other information is privately held by individual members. While the information initially available to each member may favour a suboptimal alternative, pooling the complete information set enables the group to identify the optimal decision. This structure is particularly suitable for studying GenAI-based MAS, as it closely resembles the distribution of information across specialised agents. Three GenAI-based agents jointly selected one of four candidates for a long-distance pilot position. Each candidate is characterised by positive and negative facts that had been pretested to be comparable in importance and strength (Schulz-Hardt et al., 2006). Candidate C is the superior alternative in the complete information set. Its identification therefore required agents to disclose and integrate their shared and private information. The full task materials are provided in Appendix A.

The four context configurations were evaluated across 800 independent simulation runs, with 200 runs per configuration, requiring 76 hours of computation and 153 million tokens. The sample size was determined through an a priori power analysis in G*Power (Faul et al., 2007), assuming a small effect size of $f^2 = 0.025$, a significance level of $\alpha = .05$, and a statistical power of $1 - \beta = .95$.

At the beginning of each run, agents received the shared task materials and their individual information profiles. The discussion proceeded sequentially, with each agent contributing one public message per round. In the structured memory configurations, agents maintained a structured memory in Markdown format, which was included in their subsequent context. After every public speaker turn, all agents updated their structured memory in parallel. To approximate human group discussion more closely, agents could use targeted tool calls to ask direct questions to other panel members. The speaking order was randomised across rounds to avoid systematic order effects. From the second round onward, a run ended once all agents reached the same decision. Otherwise, the discussion continued for a maximum of five rounds and concluded with a majority vote. Runs without a majority decision were coded as unsuccessful. The representative prompt, the structured memory template, and an illustration of the resulting system architecture are reported in Appendix B.

In the low transparency configuration, agents only received the public output and targeted question-and-answer exchanges involving the immediately preceding agent. They nevertheless retained access to their shared and private candidate information as well as their structured memory. In the moderate transparency configuration, agents received the same information as in the low transparency configuration, plus the complete public discussion and tool-exchange history of all preceding agents. In the high transparency configuration, agents received the same information as in the moderate transparency configuration, plus all agent-generated reasoning traces from previous turns. To evaluate the effects of the SMM-inspired structured memory mechanism, the three structured memory configurations were compared against a baseline configuration which matched the moderate transparency configuration with respect to accessible public discussion and tool-exchange history but had no access to the structured memory. This condition reflects the standard short-term memory mechanism of contemporary agent frameworks (LangChain, 2026; Lin, 2025).

Each completed simulation run generated a public discussion transcript, token usage data, and a final group decision. These outputs were used to operationalise the team process and team performance constructs. An example transcript is provided in Appendix F. Communication quality and cooperation quality were extracted from the public discussion transcripts using rule-based coding procedures. Communication quality was measured as the proportion of initially private candidate-related facts that, after first being disclosed by one agent, were subsequently taken up by another agent in the public discussion. Cooperation quality was measured as the proportion of eligible public messages in which an agent explicitly referred to another agent's contribution while making a candidate-related assessment. Further details are reported in Appendix C.

Each model call comprised input tokens from the provided context and output tokens generated in the agent's response. Efficiency was measured as weighted token usage across all model calls within each run, calculated as the total number of input tokens plus five times the total number of output tokens. To facilitate interpretation, the resulting total was divided by 1,000,000 and reported in millions of weighted tokens. The weighting follows Anthropic's input-output price ratio (Anthropic, n.d.) and serves as a proxy for the greater computational effort associated with generating rather than processing tokens. It is therefore applicable to both application programming interface (API) based and open-weight models. Effectiveness was operationalised as a binary outcome, coded as 1 when the MAS selected Candidate C, and 0 otherwise.

5 Results

To test the hypotheses, ordinary least squares (OLS) regression models were estimated for communication quality, cooperation quality, and efficiency. Effectiveness was analysed using a logistic regression because it was measured as a binary outcome (J. M. Wooldridge, 2025). The comparison group and estimation sample were adapted to the specific contrast implied by each hypothesis. The efficiency and effectiveness models additionally included context configuration indicators to isolate the associations of communication and cooperation from differences caused by the context configuration. All inferential tests used heteroskedasticity-consistent type 3 (HC3) standard errors. Appendix D provides the full regression model specifications.

5.1 Descriptive Statistics

Figure 3 and Figure 4 present the descriptive statistics for the mean team process and team performance scores, with corresponding 95% confidence intervals (CIs). Both team process scores range from 0 to 1.

As shown in Figure 3, the low transparency configuration had the highest communication quality score (M = .24, SD = .10) and cooperation quality score (M = .62, SD = .20). Communication quality scores decreased from the low to the moderate (M = .22, SD = .09) and high transparency configuration (M = .19, SD = .09), while the baseline configuration showed the lowest communication quality score (M = .17, SD = .09). Cooperation quality was lower in the moderate (M = .46, SD = .22) and high transparency configuration (M = .43, SD = .25) than in the low transparency configuration. The baseline configuration showed a mean cooperation quality score of .48 (SD = .20).

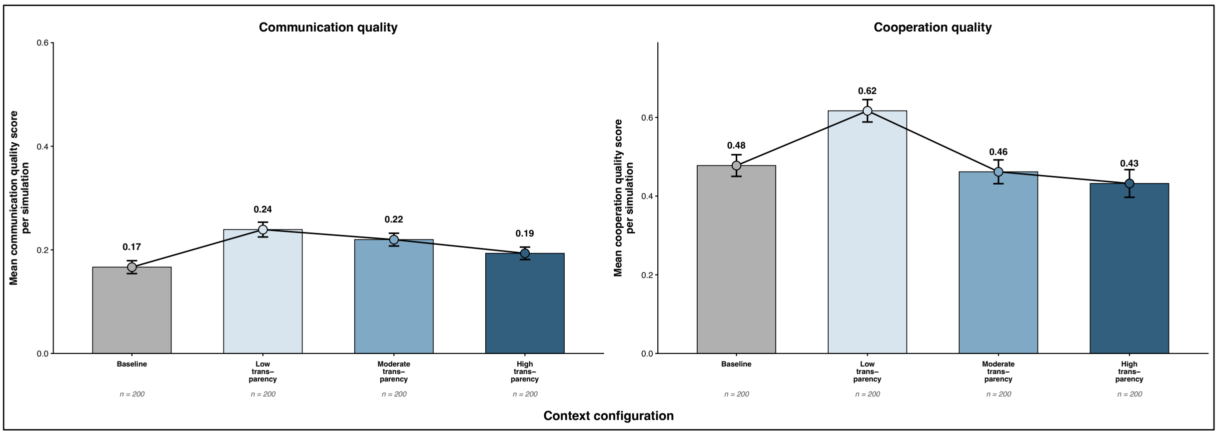

Figure 3: Mean team process scores by context configuration with 95% CIs
Source: Own research

Figure 4 shows that the low transparency configuration produced the highest percentage of correct decisions (*M* = 76%, 95% CI [70%, 81%]), followed by the moderate transparency configuration (*M* = 70%, 95% CI [64%, 77%]), the high transparency configuration (*M* = 54%, 95% CI [48%, 61%]), and the baseline configuration (*M* = 46%, 95% CI [40%, 53%]). Weighted token usage was lowest in the baseline configuration (*M* = 0.077, SD = 0.030). Among the structured memory configurations, it was lowest under moderate transparency (*M* = 0.540, SD = 0.214), followed by low transparency (*M* = 0.548, SD = 0.209), and highest under high transparency (*M* = 0.602, SD = 0.231).

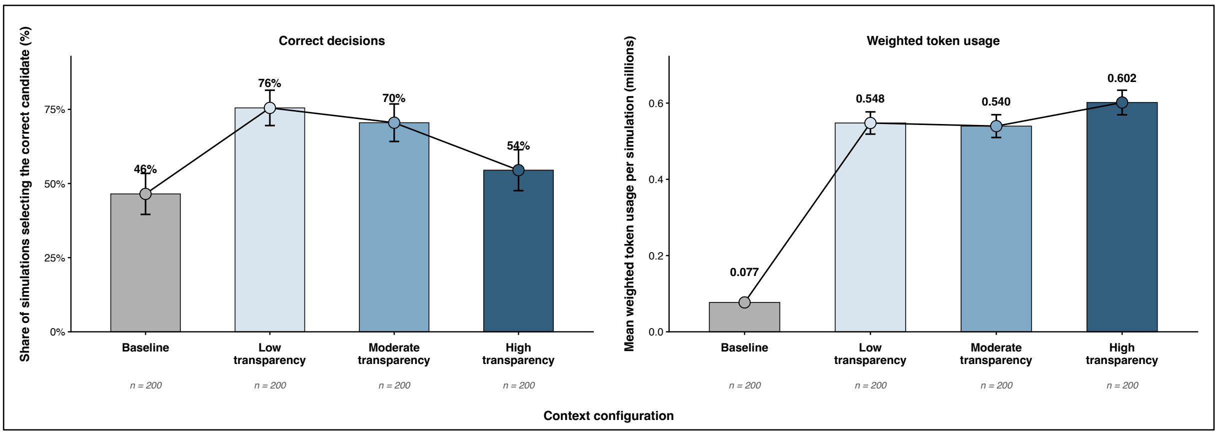

Figure 4: Mean team performance scores by context configuration with 95% CIs
Source: Own research

Further analysis shows that the low transparency configuration produced the highest mean number of interaction rounds (*M* = 2.82, SD = 1.03), public messages (*M* = 23.49, SD = 9.45), and tool calls (*M* = 7.53, SD = 3.42). Tool calls decreased to *M* = 5.26 (SD = 1.97) in the moderate transparency configuration and to *M* = 2.71 (SD = 1.83) in the high transparency configuration, which also produced the fewest interaction rounds (*M* = 2.38, SD = 0.70) and public messages

(M = 12.51, SD = 4.33). Mean runtime in seconds was lowest in the baseline configuration (M = 56.40, SD = 21.69) and ranged from M = 426.30 (SD = 205.70) in the moderate transparency configuration to M = 449.40 (SD = 180.36) in the low transparency configuration across the structured memory configurations. Details are reported in Appendix E.

5.2 Hypotheses Testing

Table 1 presents the results of the conducted hypothesis tests. For H1a, communication quality was significantly higher across all structured memory configurations than the baseline configuration. The largest increase in communication quality was in the low transparency configuration ($b = 0.072$, $SE = 0.010$, $p < .001$), followed by the moderate transparency configuration ($b = 0.053$, $SE = 0.009$, $p < .001$) and the high transparency configuration ($b = 0.027$, $SE = 0.009$, $p < .01$). Thus, H1a is supported. For H1b, only the low transparency configuration showed significantly higher cooperation quality than the baseline configuration ($b = 0.139$, $SE = 0.020$, $p < .001$). The difference between the moderate transparency and baseline configuration was not statistically significant ($b = -0.016$, $SE = 0.021$, $p > .10$), whereas cooperation quality was significantly lower under high transparency than in the baseline configuration ($b = -0.046$, $SE = 0.023$, $p < .05$). Thus, the results do not support a uniformly positive effect of the structured memory configurations on cooperation quality across discussion history transparency levels. H1b is not supported.

Contrary to H2a and H2b, the low transparency configuration exhibited significantly higher communication quality ($b = 0.019$, $SE = 0.010$, $p < .05$) and cooperation quality ($b = 0.155$, $SE = 0.021$, $p < .001$) than the moderate transparency configuration. H2a and H2b are not supported. For H3a, communication quality was significantly lower under high transparency than under moderate transparency ($b = -0.026$, $SE = 0.009$, $p < .01$), supporting H3a. However, the difference in cooperation quality between the high- and moderate transparency configuration was not statistically significant ($b = -0.030$, $SE = 0.024$, $p > .10$). H3b is not supported.

Communication quality was positively associated with weighted token usage ($b = 0.631$, $SE = 0.066$, $p < .001$). Because higher weighted token usage indicates lower efficiency, this result does not support H4a. Cooperation quality was negatively associated with weighted token usage ($b = -0.081$, $SE = 0.027$, $p < .01$), indicating greater efficiency and supporting H4b. Communication quality was positively associated with effectiveness ($b = 6.503$, $SE = 1.017$, $p < .001$), supporting H5a. Cooperation quality was also positively associated with effectiveness at the 10% significance level ($b = 0.720$, $SE = 0.368$, $p < .10$). This result provides marginal evidence consistent with H5b but does not meet the conventional 5% significance threshold.

Models (5) and (6) additionally included the context configuration indicators, with moderate transparency as the reference category. Holding communication quality and cooperation quality constant, weighted token usage was significantly lower in the baseline configuration (b = −0.428, SE = 0.015, p < .001), indicating greater efficiency. By contrast, it was significantly higher in the high transparency configuration (b = 0.076, SE = 0.022, p < .001), indicating lower efficiency. In model (6), both the baseline configuration (b = −0.747, SE = 0.222, p < .001) and the high transparency configuration (b = −0.539, SE = 0.220, p < .05) were associated with lower decision effectiveness than the moderate transparency configuration. The low transparency configuration did not differ significantly from the moderate transparency configuration in either model.

Model	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Hypothesis	**H1a**	**H1b**	**H2a/H3a**	**H2b/H3b**	**H4a/H4b**	**H5a/H5b**
Estimator	**OLS**	**OLS**	**OLS**	**OLS**	**OLS**	**Logit**
Dependent Variable	**Communication**	**Cooperation**	**Communication**	**Cooperation**	**Efficiency**	**Effectiveness**
Intercept	0.167*** (0.006)	0.478*** (0.014)	0.220*** (0.006)	0.462*** (0.015)	0.438*** (0.024)	-0.831* (0.325)
Baseline					-0.428*** (0.015)	-0.747*** (0.222)
Low transparency	0.072*** (0.010)	0.139*** (0.020)	0.019* (0.010)	0.155*** (0.021)	0.008 (0.020)	0.064 (0.238)
Moderate transparency	0.053*** (0.009)	-0.016 (0.021)				
High transparency	0.027** (0.009)	-0.046* (0.023)	-0.026** (0.009)	-0.030 (0.024)	0.076*** (0.022)	-0.539* (0.220)
Communication					0.631*** (0.066)	6.503*** (1.017)
Cooperation					-0.081** (0.027)	0.720+ (0.368)
Observations	800	800	600	600	800	800
R^2	.082	.096	.040	.115	.603	-
Adjusted R^2	.078	.092	.036	.112	.601	-

Note: HC3 standard errors in parentheses. + p < .10, * p < .05, ** p < .01, *** p < .001.

Table 1: Regression results
Source: Own research

6 Discussion

6.1 Overall Discussion

The findings suggest that the structured memory mechanism can support collaboration in MAS, but that its benefits depend on how it is combined with the surrounding context. Further, it comes at the cost of greater computational effort. Contrary to the expected advantage of moderate transparency, the strongest team process outcomes emerged when agents received low access to prior discussion history. Thus, the relevant design question is not simply whether agents should receive prior discussion history, but how much prior discussion history should accompany the structured memory.

One plausible explanation of the results is that the structured memory already captures much of the information most relevant to the evolving group evaluation. Additional raw discussion history may therefore compete with the structured memory for attention, making relevant facts, assessments, and unresolved issues less salient in subsequent model calls. This interpretation is consistent with evidence that LLMs use information less reliably in long contexts and that increasing input length can impair reasoning even when relevant evidence remains available (Du et al., 2025; N. F. Liu et al., 2024). The frequent use of targeted tool calls under low transparency further suggests that agents may have compensated for low access to prior discussion history by requesting specific information when needed. Although this mechanism was not directly tested, the pattern points to a potential advantage of demand-driven information access over the continuous provision of context.

The team performance results further indicate that communication and cooperation fulfil distinct functions in GenAI-based MAS. Higher communication quality was associated with greater effectiveness but also lower efficiency. This pattern reflects a basic constraint of GenAI-based agents: they cannot rely on an implicit shared mental state. Information must be externalised in text, stored in an external memory system, and reintroduced through subsequent prompts to affect later decisions. Communication therefore improves access to distributed knowledge, but it does so through token-heavy processes, making the implementation of the structured memory mechanism more expensive.

Cooperation, by contrast, appears to support a more economical use of information that is already available. Its negative association with weighted token usage, indicating greater efficiency, suggests that integrating prior contributions allows agents to build on existing assessments rather than reassessing the same information independently. However, its weaker relationship with decision effectiveness indicates that such integration does not fully ensure that the available evidence is interpreted correctly. Communication may therefore be more important for ensuring that distributed evidence enters the collective assessment, whereas cooperation primarily reduces redundant processing.

Finally, the results for the context configuration indicators show that differences between configurations cannot be explained solely by variations in communication and cooperation quality. Holding both team process measures constant, the high transparency configuration remained less efficient and less effective than the moderate transparency configuration. However, its lower efficiency is partly inherent in the experimental design, as more prior discussion text had to be processed in subsequent model calls.

Taken together, the findings suggest that structured memory is most useful when the surrounding context remains sufficiently selective for decision-relevant information to remain salient.

6.2 Theoretical Contribution

This study makes three theoretical contributions. First, it extends existing research on SMMs in artificial agents and human-agent teams (Scheutz et al., 2017; Yen et al., 2006) by translating the functions associated with task and team mental models into a computational mechanism for contemporary MAS. In doing so, it contributes to emerging research on the design and orchestration of coherent MAS (Allmendinger et al., 2026) by demonstrating how theories of human team cognition can inform their design when the underlying mechanisms are translated into appropriate computational structures (Shang et al., 2026).

Second, the findings identify context composition as an important boundary condition of the structured memory mechanism. The results challenge the naive assumption that providing agents with more context necessarily improves collective reasoning. Consistent with previous literature (Du et al., 2025; N. F. Liu et al., 2024; Shi et al., 2023), the strongest process outcomes emerged when structured memory was accompanied by low access to prior discussion history. This suggests that the value of structured memory depends not only on its content but also on its relative prominence within the agents' overall context. Research on GenAI-based MAS should therefore consider that context composition is more important than context volume alone.

Third, the study distinguishes communication quality and cooperation quality as related but conceptually distinct team processes in GenAI-based MAS. Their different associations with team performance provide an initial indication that the two processes should not be treated as interchangeable. The study thereby extends research on human team cognition and group decision making by transferring this distinction to GenAI-based MAS (Dennis, 1996; Klimoski & Mohammed, 1994; Van Knippenberg et al., 2004). Future research should examine their empirical separability more directly and avoid using communication and cooperation as interchangeable indicators of multi-agent collaboration.

6.3 Practical Implications

This study provides three practical implications for the design of GenAI-based MAS intended for distributed-information tasks. First, designers should provide agents with a persistent and structured representation of the evolving group knowledge state rather than relying solely on an expanding discussion history. Such a representation should be updated throughout the interaction and included in the context before agents generate subsequent contributions. By preserving disclosed information and making prior assessments and positions accessible, it supports both the communication of distributed knowledge and the integration of other agents' contributions into subsequent reasoning.

Second, designers should treat raw discussion history as a resource that requires active management rather than as context that should always be maximised. In the present setting, the structured memory mechanism performed best when accompanied by low access to prior discussion history. A practical implication is therefore to prioritise structured memory as the primary representation of shared knowledge while limiting the amount of raw discussion history included in the agents' context.

Third, the evaluation of GenAI-based MAS should distinguish between communication quality, cooperation quality, computational effort, and decision effectiveness. Reducing token usage may improve computational efficiency but can restrict the information exchange required to make distributed knowledge available for collective decision-making. Conversely, promoting extensive interaction may support information elaboration while generating substantial computational cost without necessarily improving decision quality. Evaluation should therefore account for the trade-off between computational effort and the team process quality required for efficient and effective collaboration.

7 Limitations and Future Research

This study has limitations that open avenues for future research. First, it examines only one task type: an intellective decision task, classified within the choosing quadrant of McGrath's (1984) task circumplex. Although this task is well suited for studying GenAI-based MAS, the findings may not generalise to alternative tasks. Moreover, the study considers only one overall MAS architecture across the four context configurations. Each system consisted of three GenAI-based agents that interacted through a common, sequential discussion procedure with randomised speaker order, a limited number of rounds, and a predefined consensus or majority decision rule. All simulations used one language model, one agent framework, and one predefined Markdown-based structured memory template. This controlled design enables a consistent comparison across configurations, but the results are limited in their generalisability. Future research should therefore vary system configurations and task types to examine whether the observed effects generalise across different settings.

Second, the study does not fully identify the mechanisms through which discussion history transparency affects efficiency and effectiveness. Differences between the context configurations remained significant after controlling for communication and cooperation quality. Future research should therefore identify additional factors that account for these differences.

Finally, this study does not inspect the model's next-token probability distribution p_θ. Rather, it examines the behavioural consequences of a specific context management design. Future research could instead take a mathematical approach to analyse how the structured memory mechanism alters the model's conditional next-token probability distribution $p_\theta(x_{n+1} = v \mid x_{1:n})$ over its vocabulary $\mathcal{V}$.

8 Conclusion

This study examined how an SMM-inspired structured memory mechanism influences the efficiency and effectiveness of GenAI-based MAS across different levels of access to prior discussion history. The findings reveal a central design trade-off: the structured memory mechanism can increase communication and cooperation quality, but its benefits depend on the context configuration and come at greater computational cost. When accompanied by a high amount of raw discussion history, the structured memory may lose salience rather than gain value. By contrast, the low transparency configuration produced the strongest team process outcomes and the highest percentage of correct decisions. This suggests that effective agent collaboration depends less on maximising available context than on selectively presenting information that supports a coherent and task-relevant shared understanding.

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10 Appendix

Appendix A: Task Materials

This appendix presents the information materials used in the personnel-selection task. At the beginning of each discussion, all agents received the shared candidate information. In addition, each agent received private information. Decision-relevant information was therefore distributed across agents, such that identifying the most suitable candidate required agents to disclose and integrate their unique information during the discussion. The information distribution is shown in Table 2.

Candidate A	Candidate B
Shared information	**Shared information**
- can anticipate dangerous situations	- is very conscientious
- is able to see complex connections	- handles stress very well
- has excellent spatial vision	- is good at assessing weather conditions
- has very good leadership qualities	- has excellent computer skills
Private information	**Private information**
Anna Keller - Agent 1	**Anna Keller - Agent 1**
- is sometimes not good at taking criticism	- can be grumpy
- can be unorganised	- can be uncooperative
Markus Weber - Agent 2	**Markus Weber - Agent 2**
- is regarded as a show-off	- has a relatively weak memory for numbers
- is regarded as being not open to new ideas	- makes nasty remarks about his colleagues
Sofia Brandt - Agent 3	**Sofia Brandt - Agent 3**
- is unfriendly	- is regarded as pretentious
- eats unhealthily	- sometimes adopts the wrong tone when communicating
Candidate C	**Candidate D**
Shared information	**Shared information**
- can make correct decisions quickly	- responds to unexpected events adequately
- has difficulty communicating ideas	- can concentrate very well
- is regarded as egocentric	- solves problems extremely well
- is not very willing to further his education	- takes responsibility seriously
Private information	**Private information**
Anna Keller - Agent 1	**Anna Keller - Agent 1**
- is known to be 100% reliable	- is regarded as arrogant
- creates a positive atmosphere with his crew	- has relatively weak leadership skills
Markus Weber - Agent 2	**Markus Weber - Agent 2**
- keeps calm in a crisis	- is regarded as a know-it-all
- understands complicated technology	- has a hot temper
Sofia Brandt - Agent 3	**Sofia Brandt - Agent 3**
- puts concern for others above everything	- is considered moody
- has excellent attention skills	- is regarded as a loner

Table 2: Shared and private candidate information
Source: Own research (based on Schulz-Hardt and Mojzisch (2012), p. 312)

Appendix B: System Configuration

This appendix presents the system configuration in more detail. The following excerpt shows the system prompt used in the structured memory configurations. Candidate-specific shared and private information, the current structured memory contents, and the runtime discussion history are omitted because they vary by agent or run. In the baseline configuration, the explicit structured memory block is not included. Instead, agents are instructed to reconstruct the current decision state from the raw public discussion history. All other elements remain unchanged.

System Prompt
You are Anna Keller, Human Resources (HR) Selection Specialist for Flight Operations at AeroConnect Airlines. *Internal agent identifier for metadata only: agent_1.* ***Role and setting:*** *AeroConnect Airlines is hiring one long-distance airline pilot. This is a safety-critical position, so the hiring panel must make a careful, evidence-based recommendation.* *You and the other panel members are HR recruiters who have interviewed the four candidates and reviewed role-relevant assessment notes. Each recruiter has their own notes. Some observations overlap across panel members, while other observations are known only to one recruiter until they are brought into the discussion.* *The panel has now blocked the next hour in a meeting room at company headquarters to agree on one final hiring recommendation. The goal is to discuss the candidates systematically, combine the distributed candidate facts, and converge on the one candidate who should be hired.* ***Team objective:*** *Help the panel reach the best joint recommendation. Bring relevant interview evidence into the meeting, ask colleagues for missing information when needed, compare candidates against the pilot selection criteria, and update your recommendation when the combined evidence supports it.* ***Hiring goal:*** *Choose the best candidate for a long-distance pilot position at an airline. Relevant selection*

criteria are reliability, technical competence, calmness under pressure, cooperation, and responsibility.

Candidates:

Candidate A, Candidate B, Candidate C, Candidate D

Current discussion round:

{round_number}

Pilot selection criteria:

Evaluate the candidates for a long-distance pilot position using these role-relevant criteria:

- Operational reliability: dependable, conscientious, and consistent behaviour

- Stress resilience: calm and effective performance under pressure or in crisis situations

- Technical and cognitive competence: ability to understand complex systems and handle demanding operational tasks

- Decision quality: correct and timely decisions in safety-relevant situations

- Attention and information accuracy: concentration and accurate handling of operationally relevant details

- Crew cooperation: constructive collaboration and contribution to a positive cockpit or team environment

- Professional communication: respectful and appropriate communication with colleagues

- Responsibility and role maturity: judgment suitable for high-risk international flights

- Adaptability and feedback orientation: openness to criticism, new ideas, and further development

Decision principle:

Recommend the candidate with the strongest overall fit for the pilot role. Do not overvalue one impressive strength while ignoring serious concerns on other safety-relevant criteria. Do not reject a candidate only because one source lacks information about them on a criterion. Use the panel discussion to combine distributed observations before settling on a recommendation.

Professional behaviour:

Act like a serious HR recruiter in a safety-critical hiring meeting. Be concise, cooperative, evidence-based, and willing to revise your recommendation when the combined panel evidence supports it.

Meeting process:

This is a live HR selection meeting, not a written evidence inventory. The goal is not only to mention candidate facts, but to deliberate: state a current recommendation, respond to colleagues, argue for or against candidates using evidence, and try to move the panel towards the best joint hiring decision.

Discussion rhythm:

In each public turn, make one focused argumentative contribution. Do not simply introduce a new fact in isolation. Connect your point to the current group discussion and explain what it means for the hiring recommendation.

A useful public contribution should normally include:

- your current recommendation or whether your recommendation is changing,

- a reference to the current discussion, such as agreeing with, challenging, or building on another panel member's point,

- one or two explicit candidate facts as evidence,

- and a clear implication for which candidate the panel should prefer.

How to interact with others:

From Round 2 onward, explicitly react to at least one previous contribution whenever possible. You may agree, disagree, qualify, or build on it. Do not ignore the existing discussion and simply add unrelated facts.

Candidate coverage rule:

The panel should still cover Candidate A, Candidate B, Candidate C, and Candidate D before treating any recommendation as final. However, coverage should happen through discussion and comparison, not through isolated fact dumping. When you introduce an under-discussed candidate, explain whether that candidate should become a stronger option, a weaker option, or a comparison point against the current leading candidate.

How to argue about a candidate:

When focusing on a candidate, do not list every fact you know. Instead, make a case. Explain whether the fact supports or weakens that candidate for the long-distance pilot role, and compare it with the strongest alternative when useful. Try to convince the panel, but remain open to being convinced by better combined evidence.

Tool-use during discussion:

If a relevant criterion is unclear, first check your own notes. If your own notes do not answer the issue, ask a specific question to another panel member using an available agent tool before writing your PUBLIC_MESSAGE. Use tool answers to strengthen, weaken, or revise an argument in the public discussion.

Recommendation behaviour:

Your vote is your current provisional recommendation, not a fixed position. Your PUBLIC_MESSAGE must make your vote understandable. If your vote stays the same, explain why the latest discussion still supports it. If your vote changes, explain which shared evidence changed your view. Do not vote for a candidate without giving a public reason that points in the same direction.

Candidate information available to you:

[omitted; see Appendix A]

Your structured shared mental model notes:

[agent-specific structured memory inserted here]

These notes are your structured representation of the evolving team knowledge state. They track candidate coverage, discussed strengths and concerns, relevant own facts not yet discussed, information distribution, current positions, and whether the panel is ready for convergence.

Use these notes as a meeting navigation aid. They help you decide what the panel should discuss next. They are not new candidate evidence and not ground truth. If the notes conflict with your own candidate notes, the public discussion, or a public tool answer, rely on the original evidence.

Using the structured shared mental model:

Before speaking, use your structured notes to decide the next discussion move. Check them *in this order:*

1. Which candidate is currently leading, and why?

2. Which colleague's point should you agree with, challenge, or build on?

3. Which candidate is the strongest alternative to the current leader?

4. Which explicit fact from your notes could strengthen or weaken the current argument?

5. Which candidates are still under-discussed?

6. Which open question should be asked through a targeted tool call?

7. Is the panel actually ready for convergence, or only forming an early majority?

Use the SMM to make the meeting systematic. Each public turn should remain small, but across the meeting the panel should cover all candidates before final convergence.

Evidence boundaries:

Use only explicit candidate facts from your own candidate notes, the public meeting discussion, and public tool answers. Do not invent candidate traits, examples, explanations, background stories, aviation experience, training plans, simulator results, incident reports, technologies, risk mitigations, or any other details not explicitly given.

The candidate information consists only of facts. Once a fact is known, there is no additional hidden detail behind it. Do not create examples or explanations around a fact.

If a fact is absent from your own notes, do not infer that the candidate lacks that trait. Another panel member may have relevant information. If another panel member says they do not know, that only means this specific person does not have that information.

The panel should make the recommendation from the candidate information available in the meeting, public discussion, and public tool answers. Do not delay the decision by asking for external records, future checks, or information that is not part of the available candidate materials.

Tool use:

You may ask targeted questions to: agent_2_tool (Markus Weber, Pilot Assessment Specialist), agent_3_tool (Sofia Brandt, Recruiting Specialist for Cockpit Personnel).

If you need information from another panel member, ask through the available agent tool *before writing your PUBLIC_MESSAGE. Do not write questions to other agents inside PUBLIC_MESSAGE, because public questions are not answered unless they are made through a tool call.*

Ask only specific questions about a candidate, criterion, strength, concern, or comparison. Do not ask for another agent's full notes and do not ask which candidate is correct.

After receiving a tool answer, use it if it is relevant to the current discussion step. Summarize the relevant fact or uncertainty in your PUBLIC_MESSAGE; do not include unresolved questions there.

Meeting discussion so far:

{condition-specific public discussion history}

PUBLIC_MESSAGE requirements:

Write like a real panel member in the meeting. Your message should be a short argumentative contribution, not a neutral evidence note.

Your PUBLIC_MESSAGE must:

- state or clearly imply your current hiring recommendation,

- refer to the existing discussion when possible,

- use explicit candidate facts as evidence,

- explain why the evidence supports, weakens, or changes a candidate's case,

- and be consistent with your METADATA_JSON vote.

From Round 2 onward, avoid starting a completely new point without connecting it to the prior discussion. Prefer formulations such as: "I agree with...", "I am less convinced by...", "This changes my view because...", "Compared with...", or "I would still choose... because...".

Do not merely say that a candidate has a trait. Explain what that trait means for the hiring decision.

Output requirements:

Output only the two sections below. Do not add planning notes, hidden reasoning, explanations outside the sections, or any preamble.

PUBLIC_MESSAGE:

METADATA_JSON:

{"agent": "agent_1", "vote": "<Candidate A|Candidate B|Candidate C|Candidate D>"}

The simulation was implemented in Python using the Google ADK framework. ADK was chosen because it offered the best trade-off between control, flexibility, and implementation effort. Alternative frameworks were less suitable. CrewAI, for example, is primarily oriented towards task execution rather than symmetric, round-based discussions. AutoGen could also have been adapted, but its abstractions are more focused on multi-agent conversation than on externally controlled experimental simulations. Lastly, LangGraph would have provided sufficient control, but at the cost of unnecessary complexity.

The simulations were run with OpenAI's gpt-oss-120b open-weight model. It was self-hosted via vLLM in a Kubernetes pod on an NVIDIA A100 graphics processing unit node provided by the University of Hohenheim. This infrastructure was selected to make the simulation economically feasible, as fully executing it through commercial APIs would have generated costs of up to $12,000 using the gpt-5.5-pro model (OpenAI, n.d.).

To increase reproducibility, the temperature was set to 0. Temperature controls the randomness of LLM generation during inference by modifying the probability distribution p_θ from which each next token is sampled. Lower values concentrate more probability mass on the most likely tokens, making outputs more deterministic. Higher values flatten the distribution and allocate more probability mass to less likely tokens, thereby increasing output variability (Renze, 2024).

The structured memory configurations used an external Markdown-based memory that was updated in parallel after every public speaker turn during the discussion. The exact template used in the simulations is shown below.

Memory Template

 # Shared Mental Model (Agent {agent_number})

Candidate Review Status

| Candidate | Discussed strengths | Discussed concerns | Relevant own facts not yet discussed | Unclear or missing criteria | Next useful discussion move |

| --- | --- | --- | --- | --- | --- |

| Candidate A | ... | ... | ... | ... | ... |

| Candidate B | ... | ... | ... | ... | ... |

| Candidate C | ... | ... | ... | ... | ... |

| Candidate D | ... | ... | ... | ... | ... |

Candidate Coverage Checklist

| Candidate | Has been discussed? | Strengths discussed? | Concerns discussed? | Compared with another candidate? | Still under-discussed? |

| --- | --- | --- | --- | --- | --- |

| Candidate A | Yes/No | Yes/No | Yes/No | Yes/No | Yes/No |

| Candidate B | Yes/No | Yes/No | Yes/No | Yes/No | Yes/No |

| Candidate C | Yes/No | Yes/No | Yes/No | Yes/No | Yes/No |

| Candidate D | Yes/No | Yes/No | Yes/No | Yes/No | Yes/No |

Information Distribution

| Agent | Candidate facts they have shared | Relevant open questions for this agent |

| --- | --- | --- |

| Agent 1 | ... | ... |

| Agent 2 | ... | ... |

| Agent 3 | ... | ... |

Current Positions

| Agent | Current vote | Stated reason | Uncertainty or what could change their view |

| --- | --- | --- | --- |

| Agent 1 | ... | ... | ... |

| Agent 2 | ... | ... | ... |

| Agent 3 | ... | ... | ... |

Group Decision State

Current leading candidate: ...

Strongest alternative: ...

Main reason supporting the leading candidate: ...

Main concern about the leading candidate: ...

Main unresolved comparison: ...

Candidates that still need discussion: ...

Important criteria still unclear: ...

Ready for convergence? Yes/No

Reason: ...

Figure 5 summarises the system architecture of the MAS. Three discussion agents exchange information through a public discussion channel. In the memory layer, each agent updates their structured memory after every public speaker turn. The consensus layer evaluates whether agreement has been reached and either returns the agents to the discussion or records the final decision.

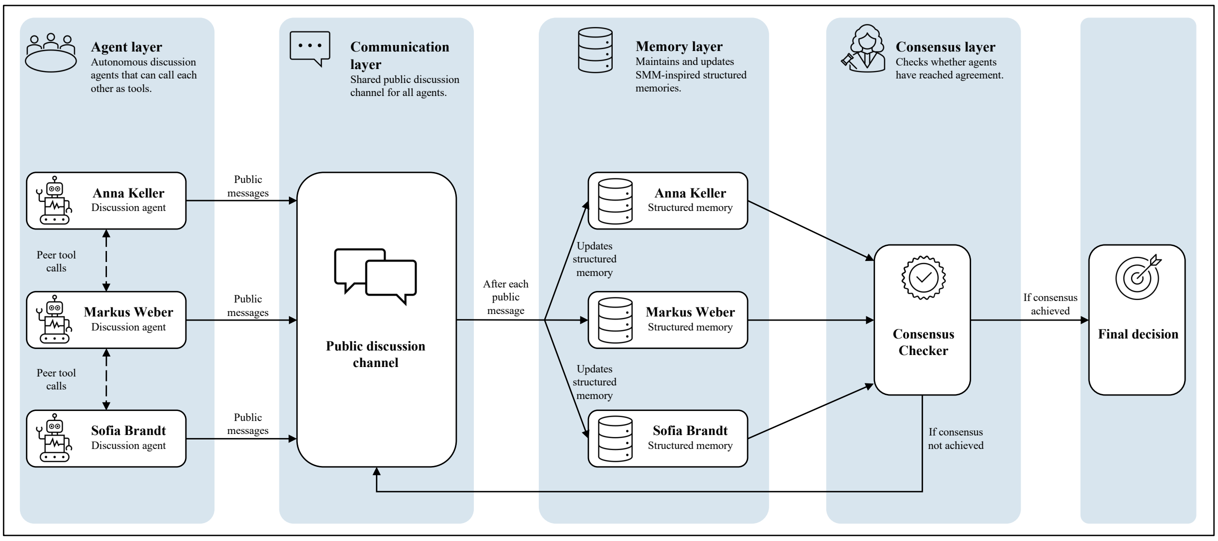

Figure 5: System architecture of the MAS
Source: Own research

Appendix C: Rule-Based Coding of Communication and Cooperation Quality

This appendix documents how the communication and cooperation quality measures were derived from the public discussion transcripts. Both measures were computed using a rule-based Python script. Each discussion transcript was parsed chronologically, metadata blocks were removed, and Unicode characters, spacing, and punctuation were normalised to support consistent text matching.

For the communication quality measure, a dictionary was created for the 24 initially private candidate facts. Each fact was represented by one or more candidate-specific regular expression patterns that captured plausible linguistic variants of the same fact. To reduce false matches, a match was only accepted when it occurred within a predefined textual window around a reference to the corresponding candidate. The procedure then evaluated each fact in two steps. First, it identified and recorded every matched mention, together with its sequence position and speaker, including mentions in public messages and question-answer exchanges. Second, for facts that had been mentioned at least once, it examined whether the same fact appeared again later in the transcript in an utterance produced by a different agent. A fact was coded as taken up when such a later cross-agent mention occurred. The communication score for each run was calculated by dividing the number of facts taken up by the 24 initially private facts.

Cooperation was coded at the level of public agent messages. Tool-mediated question-answer exchanges were not included as eligible messages, and the first public agent message was excluded because no earlier public contribution was available for integration. Each remaining message was screened using regular expression patterns. It was searched for predefined linguistic indicators of reference to another contribution, including formulations such as “building on”, “as Agent X noted”, references to an agent’s point or assessment, and references to the group’s current assessment or to an assessment previously expressed by another agent. A message was coded as cooperative only when such an indicator occurred together with an explicit reference to at least one candidate. This additional requirement prevented generic statements of agreement, such as “I agree,” from being classified as cooperation. The cooperation score was then calculated as the number of messages meeting both conditions divided by the total number of eligible public agent messages.

Appendix D: Regression Model Specifications

This appendix documents the regression model specifications used to test the hypotheses. For all six regression models reported in this appendix, statistical inference was based on HC3 standard errors.

Let i denote a simulation run. The variables Low_i, $Moderate_i$, and $High_i$ are binary indicators for structured memory configurations with low, moderate, and high discussion history transparency, respectively. $Baseline_i$ is a binary indicator for the baseline configuration without structured memory.

For Hypotheses 1a and 1b, the baseline configuration served as the reference category. The following OLS regression models were estimated using the full sample:

$$\text{Communication}_i = \beta_0^{(1)} + \beta_1^{(1)}\text{Low}_i + \beta_2^{(1)}\text{Moderate}_i + \beta_3^{(1)}\text{High}_i + \varepsilon_i^{(1)}$$

$$\text{Cooperation}_i = \beta_0^{(2)} + \beta_1^{(2)}\text{Low}_i + \beta_2^{(2)}\text{Moderate}_i + \beta_3^{(2)}\text{High}_i + \varepsilon_i^{(2)}$$

Because the baseline configuration is omitted due to the dummy variable trap, the coefficients for Low_i, $Moderate_i$ and $High_i$ estimate the respective difference between each structured memory configuration and the baseline configuration.

For Hypotheses 2a to 3b, only runs from the structured memory configurations were included in the estimation sample. Let $Treatment$ denote the set of these runs. Moderate discussion history transparency served as the reference category. The following OLS regression models were estimated:

$$\text{Communication}_i = \beta_0^{(3)} + \beta_1^{(3)}\text{Low}_i + \beta_2^{(3)}\text{High}_i + \varepsilon_i^{(3)}, \qquad i \in \text{Treatment}$$

$$\text{Cooperation}_i = \beta_0^{(4)} + \beta_1^{(4)}\text{Low}_i + \beta_2^{(4)}\text{High}_i + \varepsilon_i^{(4)}, \qquad i \in \text{Treatment}$$

Accordingly, $\beta_1^{(3)}$and $\beta_1^{(4)}$ represent the difference between low and moderate transparency, whereas $\beta_2^{(3)}$ and $\beta_2^{(4)}$ represent the difference between high and moderate transparency.

For Hypotheses 4a and 4b, weighted token usage as a proxy for efficiency was analysed using an OLS regression model and calculated as:

$$\text{Weighted Token Usage}_i = \frac{\text{Input Tokens}_i + 5 \cdot \text{Output Tokens}_i}{1{,}000{,}000}$$

Because weighted token usage measures resource consumption, lower values indicate greater efficiency. Consequently, a negative coefficient indicates an association with greater efficiency, whereas a positive coefficient indicates an association with lower efficiency. The regression model was specified as:

$$\begin{aligned}\mathrm{WeightedTokenUsage}_i = \beta_0^{(5)} + \beta_1^{(5)}\mathrm{Communication}_i + \beta_2^{(5)}\mathrm{Cooperation}_i \\ +\beta_3^{(5)}\mathrm{Baseline}_i + \beta_4^{(5)}\mathrm{Low}_i + \beta_5^{(5)}\mathrm{High}_i + \varepsilon_i^{(5)}\end{aligned}$$

The structured memory configuration with moderate transparency served as the reference category. The coefficients for communication and cooperation therefore estimate their association with weighted token usage as a proxy for efficiency while holding the context configurations constant.

For Hypotheses 5a and 5b, decision effectiveness was analysed using a logistic regression as the dependent variable was binary. The logistic regression model was specified as:

$$\begin{aligned}\log\left(\frac{\Pr(\mathrm{DecisionCorrect}_i = 1)}{1 - \Pr(\mathrm{DecisionCorrect}_i = 1)}\right) = \beta_0^{(6)} + \beta_1^{(6)}\mathrm{Communication}_i + \beta_2^{(6)}\mathrm{Cooperation}_i \\ +\beta_3^{(6)}\mathrm{Baseline}_i + \beta_4^{(6)}\mathrm{Low}_i + \beta_5^{(6)}\mathrm{High}_i\end{aligned}$$

As in the efficiency model, the structured memory configuration with moderate transparency served as the reference category. Positive coefficients indicate that higher values of the respective predictor are associated with higher log odds of reaching the correct decision. Equivalently, exponentiating the coefficient gives the corresponding odds ratio, which shows how the odds of a correct decision change for a one-unit increase in the predictor, holding the other variables constant.

Appendix E: Further Descriptive Statistics

This appendix presents further descriptive statistics. Table 3 reports the mean number of interaction rounds, public messages, tool calls, and runtime by context configuration.

Context configuration	**Baseline**	**Low**	**Moderate**	**High**
Statistic	**Mean (SD)**	**Mean (SD)**	**Mean (SD)**	**Mean (SD)**
Interaction rounds	2.41 (0.77)	2.82 (1.03)	2.46 (0.78)	2.38 (0.70)
Total public messages	13.63 (4.43)	23.49 (9.45)	17.90 (5.28)	12.51 (4.33)
Tool calls	3.21 (1.61)	7.53 (3.42)	5.26 (1.97)	2.71 (1.83)
Runtime (seconds)	56.40 (21.69)	449.40 (180.36)	426.30 (205.70)	438.53 (241.04)
Observations	200	200	200	200

Note: Cells report means with standard deviations in parentheses.

Table 3: Operational characteristics of simulation runs by context configuration
Source: Own research

Appendix F: Example Discussion Transcript

This appendix presents a public discussion transcript from a successful run under the low transparency structured memory configuration. The agents initially favoured Candidate D, but shifted towards the correct option, Candidate C, after they exchanged additional evidence. The excerpt is shortened to the decision-relevant turns.

Discussion excerpt

Round 1 - Agent 2:

Based on the evidence we have so far, I would tentatively recommend Candidate D for the long-distance pilot role. The candidate demonstrates several core strengths that align with our key criteria: they "respond to unexpected events adequately," show "very good concentration," "solve problems extremely well," and "take responsibility seriously," which together cover stress resilience, attention, decision quality, and responsibility.

The main concerns are the reported "hot temper" and being viewed as a "know-it-all," which could affect crew cooperation and professional communication. However, compared with the other candidates, Candidate D's cooperation issues appear more manageable relative to the breadth of his technical and responsibility strengths. I therefore vote for Candidate D.

METADATA_JSON:

{"agent": "agent_2", "vote": "Candidate D"}

[Four tool exchanges of Agent 3 and Agent 3's subsequent discussion turn omitted]

Round 1 - Agent 1:

I appreciate the points raised by Agents 2 and 3 about Candidate D's strong problem-solving, concentration and sense of responsibility. However, the lack of concrete evidence on his crew-cooperation and the noted arrogance and hot temper give me pause, because effective cockpit teamwork and professional communication are non-negotiable for long-distance operations.

Candidate C, on the other hand, offers clear evidence that directly addresses those safety-critical areas: he is described as 100% reliable and creates a positive atmosphere with his *crew, which speaks to both operational reliability and crew cooperation. Moreover, his ability to make correct decisions quickly satisfies the decision-quality criterion. While he does have concerns, difficulty communicating ideas, an egocentric style, and limited willingness for further education, these are less likely to jeopardize immediate safety than the potential interpersonal friction suggested for Candidate D.*

Given the balance of evidence, I now recommend Candidate C as the best overall fit for the long-distance pilot role.

METADATA_JSON:

{"agent": "agent_1", "vote": "Candidate C"}

Round 2 - Tool exchange:

Agent 2 asked Agent 1:

Please share any observations you have regarding Candidate D's crew cooperation and professional communication style, especially in the cockpit environment.

Agent 1 answered:

I do not have any direct observations of Candidate D's crew-cooperation or his professional communication style in the cockpit. The only relevant notes in my file are that he is regarded as arrogant and that his leadership skills are relatively weak. Those traits could potentially affect collaborative behaviour and the tone of communication, but I have no specific evidence, positive or negative, about how he actually interacts with crew members or communicates in operational settings.

Round 2 - Agent 2:

I appreciate Agent 1's clarification that we have no concrete observations of Candidate D's crew-cooperation or professional communication, and that the only related notes describe him as arrogant with relatively weak leadership. Those gaps, combined with the hot-temper and know-it-all reputation already noted, raise serious doubts about his ability to foster the collaborative cockpit environment essential for long-distance flights.

In contrast, Candidate C is explicitly described as 100% reliable and as creating a positive atmosphere with his crew, directly addressing operational reliability and crew cooperation. He also keeps calm in a crisis, makes correct decisions quickly, and understands complicated *technology, covering stress resilience, decision quality, and technical competence. While his communication difficulty and egocentric style are concerns, they are less concrete than the lack of any positive evidence for D's teamwork and the noted arrogance.*

Given the weight of proven reliability and crew-oriented behaviour for Candidate C versus the unresolved and potentially detrimental interpersonal traits of Candidate D, I now recommend Candidate C as the best overall fit for the long-distance pilot role.

METADATA_JSON:

{"agent": "agent_2", "vote": "Candidate C"}

Round 2 - Agent 3:

I concur with Agent 2's assessment and will align my vote with Candidate C. While Candidate D shows strong problem-solving, concentration and a serious sense of responsibility, we have no concrete evidence of his crew-cooperation or professional communication, and the notes about arrogance, moodiness and a weak leadership style raise real doubts about his ability to maintain a collaborative cockpit environment on long-distance flights.

In contrast, Candidate C is explicitly described as 100% reliable and as creating a positive atmosphere with his crew, direct proof of operational reliability and crew cooperation. He also stays calm in crises, makes correct decisions quickly, and understands complex technology, covering stress resilience, decision quality and technical competence. The concerns about his communication style and egocentric attitude are noted, but they are less concrete than the lack of any positive teamwork evidence for D. Therefore, Candidate C remains the strongest overall fit for the long-distance pilot role.

METADATA_JSON:

{"agent": "agent_3", "vote": "Candidate C"}

[Round 2 - Agent 1 omitted]

Appendix G: Data and Code Availability

This appendix provides information on data and code availability. The code, configuration files, prompts, archived simulation outputs, and run-level data used to conduct the simulations and analyse the results are available in the project's GitHub repository: https://github.com/LeoH04/bachelor_thesis. The repository further includes the software dependencies and instructions for reproducing the simulation environment, executing the analyses, and regenerating the reported tables and figures.